

CS3001 – Computer Networks

Quiz – 1A

Name	
Roll. No.	
Section	

Part 1 (3 Marks)

Provide 1-line answers to the following questions:

Why does packet switching provide better scalability for bursty traffic than circuit switching?

Packet switching shares bandwidth dynamically, letting bursty traffic use idle capacity—unlike the fixed reservations of circuit switching.

How does frequency-division multiplexing (FDM) allows multiple users to share the same physical medium without interference.

Frequency-division multiplexing assigns each user a separate frequency band, therefore these do not interfere with one another on the same medium

What is the impact of header overhead on effective throughput in a multi-layered protocol stack?

Header overhead enlarges packets, so a smaller fraction of each transmission carries payload, thereby lowering effective throughput

Part 2 (7 Marks)

Scenario

- 4 intermediate nodes, each adding **30 ms** of processing per packet.
- Packet size: **1024 bytes**.
- File to transfer: **5 000 000 bytes**.
- Minimum bandwidth of the link: **2 Mbps**.

How many **seconds** will it take to transfer the file from server to client?

Solution

STEP	CALCULATION	RESULT
Number of packets	$N = \lceil \frac{5\,000\,000}{1024} \rceil = 4883$	4883 packets
Transmission time per packet	$T_{tx} = \frac{8192}{2 \times 10^6} = 0.004096\text{s} = 4.096\text{ ms}$	4.096ms
Processing time per packet (per node)	$t_{proc} = 30\text{ ms}$	30ms
Bottleneck?	$t_{proc}(30\text{ ms}) > T_{tx} \rightarrow$ processing is the bottleneck.	–
Pipeline throughput	1 packet every 30 ms.	–
Time for first packet to finish	$T_{first} = T_{tx} + 4 \times t_{proc} = 4.096\text{ ms} + 120\text{ ms} = 124.096\text{ ms}$	124.096ms
Total transfer time	$T_{total} = T_{first} + (N - 1) \times t_{proc}$	
	$= 124.096\text{ ms} + 4882 \times 30\text{ ms}$	
	$= 124.096 + 146\,460\text{ ms} = 146\,584.096\text{ ms}$	
	$= 146.58\text{ s } (\approx 2\text{ min } 26\text{ s})$	146.6 s